# Machine Learning-Based Wind Speed Prediction in Urban Environments


**Multiscale Multiphysics Modeling and Data-Driven Analysis of Thermo-fluids Lab**

Undergraduate Student Research Award (USRA) - 2026

Yusra Kalla, Department of Electrical and Computer Engineering, Cleveland State University

Project Mentor: Dr. Navid Goudarzi, Department of Mechanical Engineering, Cleveland State University


## Background and Motivation

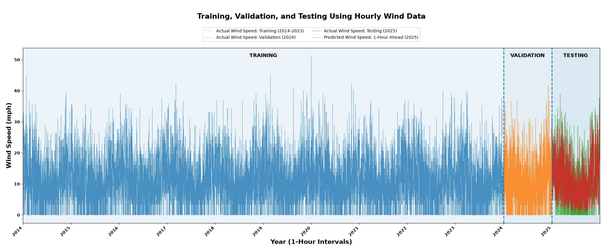
Wind speed can change quickly, making accurate short-term prediction challenging. Reliable wind-speed forecasts are important for renewable energy planning and other energy applications, especially in urban cities like Cleveland, Ohio. Machine learning can use historical weather patterns to improve these predictions. This study investigates whether machine learning models such as Artificial Neural Networks (ANNs) and XGBoost can accurately predict hourly wind speed and improve performance over a simple persistence forecast.

## Objectives

- Develop ANN and XGBoost wind-speed prediction models.
- Predict wind speed one hour ahead.
- Compare ANN and XGBoost wind-speed prediction performance with persistence.
- Evaluate performance using unseen 2025 data.

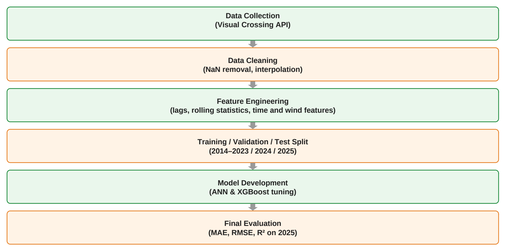
## Data

- **Location:** Cleveland, Ohio
- **Source:** Visual Crossing
- **Period:** 2014–2025
- **Resolution:** 1-hour intervals
- **Forecast horizon:** 1 hour ahead
- **Records:** 105,180




## Methodology

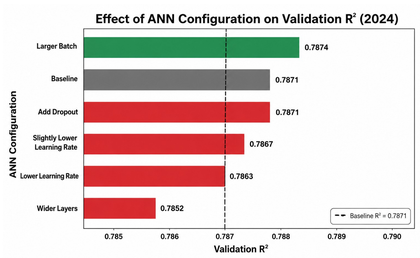
The weather data were cleaned and processed before training the models. Additional features were created to help the models recognize short-term and seasonal wind patterns.



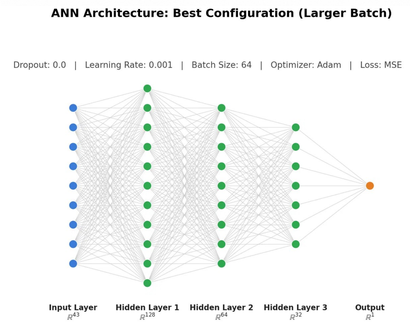

## Model development

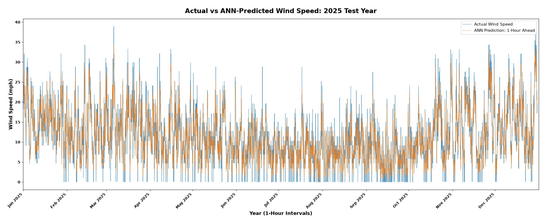
### Artificial Neural Network (ANN)

We evaluated six ANN configurations using the 2024 validation data to determine the best model architecture and training settings. The selected ANN contained three hidden layers with 128, 64, and 32 neurons, a learning rate of 0.001, and a batch size of 64. The model was trained for up to 200 epochs with early stopping to reduce overfitting. After selecting the best configuration, the final ANN was trained using the available historical training data and evaluated on the unseen 2025 test set.



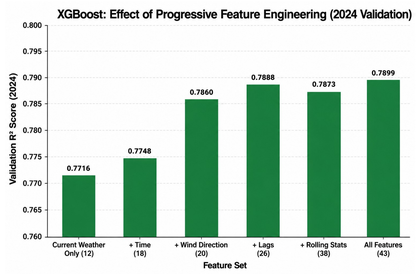

*ANN tuning produced small changes in validation performance, with the larger batch configuration achieving the highest R².*



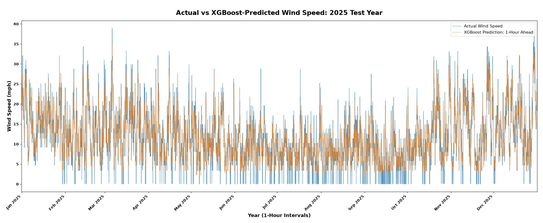




### XGBoost

XGBoost builds a sequence of decision trees, with each tree helping correct errors made by the previous trees. Different numbers of trees, tree depths, learning rates, and subsampling settings were tested using the validation data.




The selected model used 800 trees, a maximum depth of 6, and a learning rate of 0.03. Feature engineering was also evaluated to determine whether additional weather and time-based information improved prediction performance.




| Model | MSE | RMSE | $R^2$ |
|---|---|---|---|
| Persistence (baseline) | 10.7998 | 3.2863 | 0.749 |
| ANN | 8.6015 | 2.9328 | 0.7997 |
| XGBoost | 8.3089 | 2.8825 | 0.8066 |

Both machine-learning models outperformed the persistence baseline on the unseen 2025 data. XGBoost achieved the best overall performance ($R^2$ = 0.8066, RMSE = 2.8825 mph), while ANN performed closely behind ($R^2$ = 0.7997, RMSE = 2.9328 mph).

## Discussion

- Both ANN and XGBoost outperformed the persistence baseline, showing that the models learned useful patterns from historical weather data.
- XGBoost performed best overall, while ANN achieved very similar results.
- The results suggest that recent weather conditions, time-based patterns, and previous wind behavior are useful for predicting wind speed one hour ahead.
- The small performance difference between ANN and XGBoost shows that both models were effective for short-term wind-speed forecasting.




## Conclusion

XGBoost delivered the strongest one-hour-ahead wind-speed forecasting performance on unseen 2025 data ($R^2$ = 0.807), closely followed by ANN ($R^2$ = 0.800). Both machine-learning models clearly outperformed the persistence baseline ($R^2$ = 0.749), demonstrating their effectiveness for short-term wind-speed prediction.

## Acknowledgements


This research was supported by the Undergraduate Student Research Award (USRA) at Cleveland State University.

Special thanks to Dr. Navid Goudarzi, Maede Najian, and the M³FluiD Lab for their guidance and support, and Visual Crossing for giving access to local weather data.


## References


[1] Reja, R. K., Amin, R., Tasneem, Z., Abhi, S. H., Bhatti, U. A., Sarker, S. K., Ain, Q. U., & Ghadi, Y. Y. A New ANN Technique for Short-Term Wind Speed Prediction Based on SCADA System Data in Turkey.

[2] Zheng, H., & Wu, Y. (2019). *A XGBoost model with weather similarity analysis and feature engineering for short-term wind power forecasting*. Applied Sciences, 9(15), 3019.